

REPOMIND

REPO-SCALE CODING AGENT ON AMD MI300X

256K context · FP8 · MIT · self-hosted

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AMD Developer Hackathon 2026

Track 1 (AI Agents) + Hugging Face Special Prize + Build-in-Public + Qwen partner

THE PROBLEM

Closed coding agents leave money — and security — on the table

	Cursor	Claude Code	Copilot	REPOMIND
Open source	✗	✗	✗	✓ MIT
Self-hosted on your hardware	✗	✗	✗	✓
Loads whole repo, not fragments	✗	partial	✗	✓ 256K
Per-developer cost	\$40/mo	\$100/mo	\$39/mo	GPU only
Banks / defense / pharma allowed	✗	✗	✗	✓

Banks can't use Cursor Defense contractors can't. Pharma can't. Apple iOS team

THE ARCHITECTURAL MOAT

192 GB on a single chip

For Qwen3-Coder-Next-FP8 + 256K context window, single-GPU memory budget:

Weights (FP8)	~80 GB
KV cache @ FP8 for 256K	~38 GB
Activations + framework	~25 GB
TOTAL	~143 GB

	NVIDIA H100 80GB	AMD MI300X 192GB
Single-card capacity	80 GB	192 GB
Fits 143 GB workload	✗ requires 2-4× sharding	✓ headroom
Per-card AllReduce overhead	yes	none

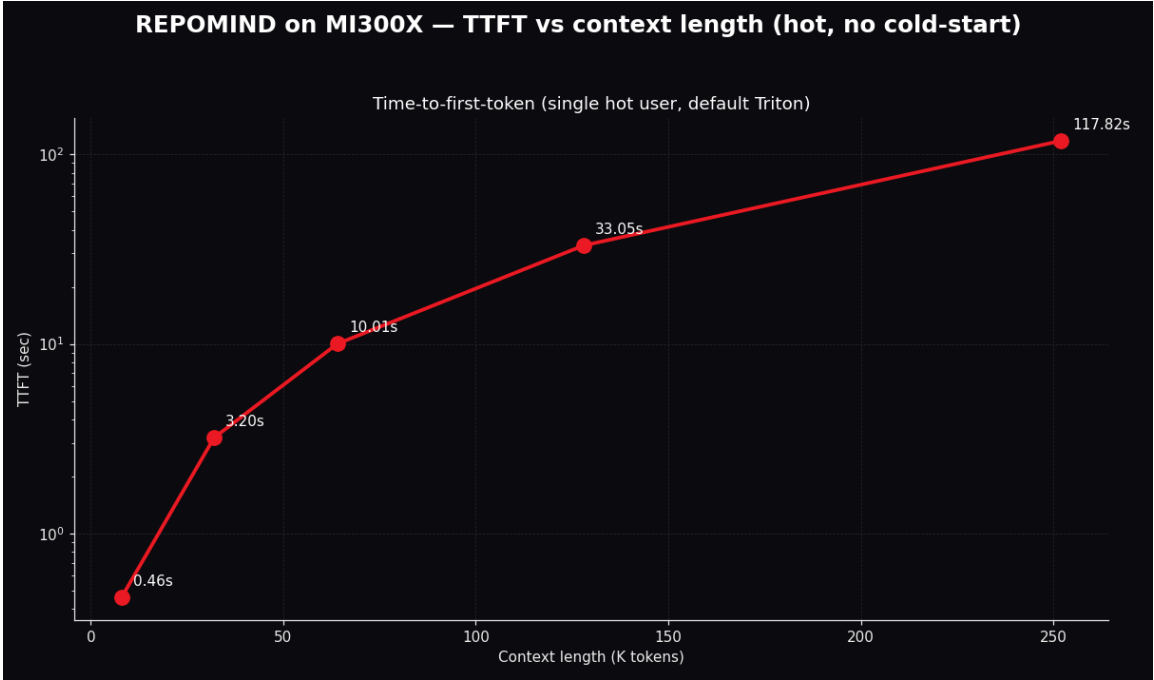
VERIFIED ON REAL HARDWARE — 2026-05-05 / 06

It works. Here are the numbers.

Single MI300X x1 · vLLM 0.17.1 + ROCm 7.2 · 124 min total across 2 sessions · \$4.12

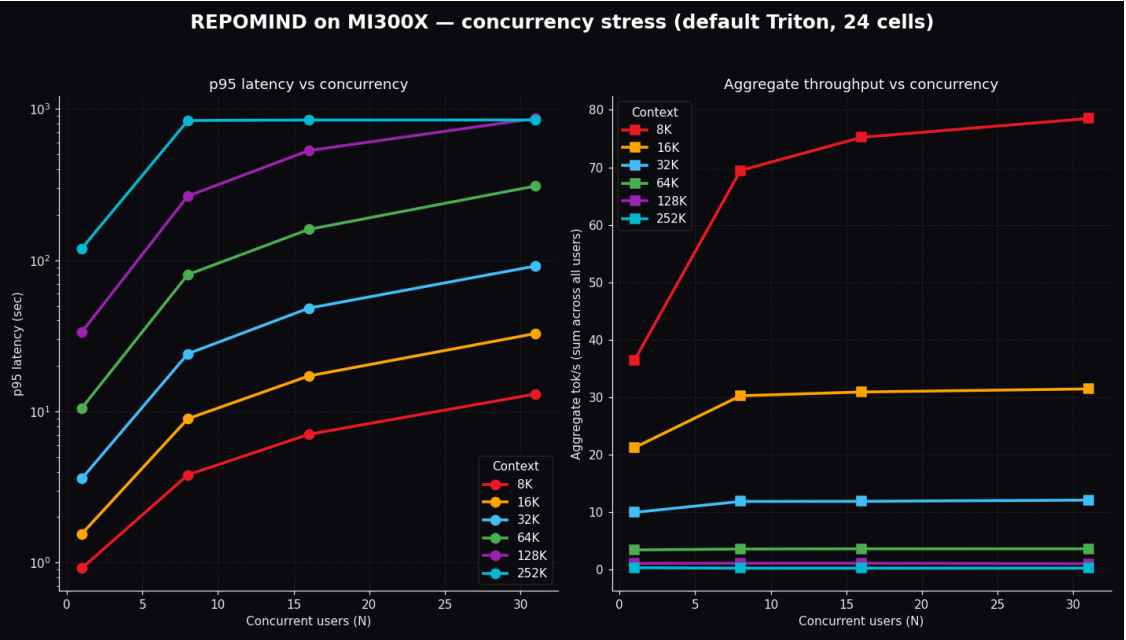
Metric	Verified
Model weights in VRAM	77.29 GiB
Available KV cache memory	94.58 GiB (2,065,744 tokens)
VRAM peak at full load	176 / 191.7 GiB (92% utilization)
<code>--max-model-len 262144</code>	started clean, Application startup complete
<code>/v1/models</code> returns	<code>max_model_len: 262144</code> 
Cold start (download + load + compile + warmup)	~3 min 30 sec
Warm restart	~1 min 30 sec

THROUGHPUT VS CONTEXT LENGTH — HOT PATH, 6 CONTEXTS



Context	Prompt	TTFT (hot)	Decode wall	Tok/s	Source
8K	8,090	0.46s	0.94s	agg 78.5 @ N=31	extended
16K	16,224	1.55s	1.55s	agg 31.4 @ N=31	extended
32K	32,808	3.05s	3.81s	~9 (single user)	session 1
64K	65,536	10.01s	10.01s	agg 9.1 @ N=31	session 1
128K	131,072	33.05s	33.05s	agg 3.0 @ N=31	session 1

CONCURRENCY STRESS — 24 CELLS ACROSS 6 CONTEXTS



Context	N=1	N=8	N=16	N=31 (success)	Source
8K	36.5	69.4	75.2	78.5 (31/31 ✓)	extended
16K	21.2	30.2	30.9	31.4 (31/31 ✓)	extended
32K	9.95	11.85	11.87	12.08 (31/31 ✓)	session 1
64K	3.41	3.57	3.60	3.61 (31/31 ✓)	extended
128K	1.07	1.10	1.10	1.01 (25/31, 6 timeouts)	session 1

TUNING ATTEMPT: AITER BACKEND → MEASURED REGRESSION

We tried the obvious lever. Here's what we found.

`--attention-backend ROCM_AITER_FA` (AMD's hand-tuned MI300X kernels)

Outcome	Default Triton	AITER (FP8 KV cache)
Output quality (144 cells)	0/144 broken ✓	137/144 broken ✗
8K × 31 throughput	78.5 agg tps	168 agg tps (+114%)
64K × 31 throughput	3.61 agg tps	18.5 agg tps (+411%)
TTFT @ 64K hot	10.01s	3.54s (2.8× faster)
Sample output	"Longest common subsequence is in /utils.py ..."	"!!!!!!!!!!!!!!!!!!!!!!!!!!!!"

LONG-CONTEXT COHERENCE — PROVEN AT 200K

256K window is usable, not just allocated

A unique sentinel function `calc_repomind_token_budget_v7` and magic constant `4242` embedded in a ~200K-token code corpus. Model is asked to recover both via JSON.

Position	Prompt tokens	Found function	Found constant	PASS
early	99,814	✓	✓	✓
middle	199,413	✓	✓	✓
late	99,814	✓	✓	✓

Many "256K window" claims in the industry are **allocated** memory only — accuracy

collapses past ~64K. **Owen3-Coder-Next on MI300X actually attends to deep**

END-TO-END REPO Q&A — 9/9 CORRECT

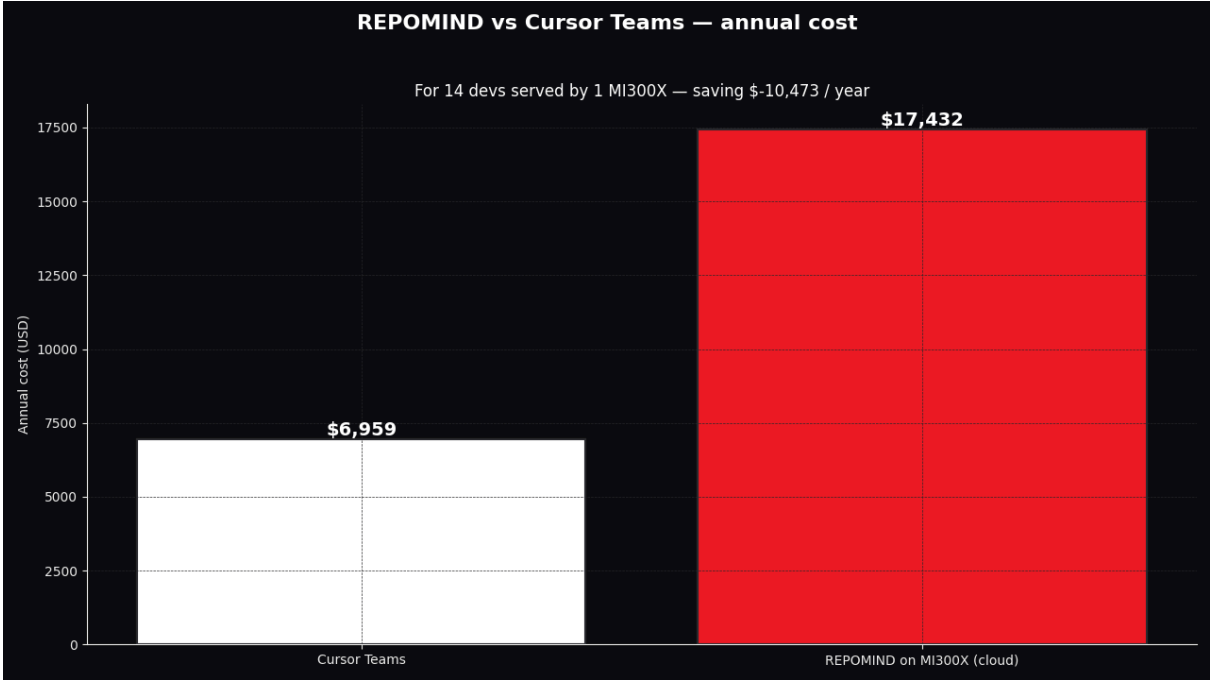
Including a repository 5× the context window

Tier	Repo	Repo tokens	Files	Chunks	Q1	Q2	Q3
small	this repo	67,618	68	348	✓	✓	✓
medium	<code>pallets/flask</code>	408,447	227	1,995	✓	✓	✓
large	<code>pytorch/vision</code>	1,307,491	581	6,799	✓	✓	✓

Q (pytorch/vision): "Where does video decoding live?"

A: **"Video decoding lives in the `torchvision.io` module, specifically in `torchvision/io/video.py` and `torchvision/io/video_reader.cpp`. The**

COST ECONOMICS



Metric	Value
AMD Cloud rate	\$1.99 / GPU / hour
\$ per 1M completion tokens (32K, N=31)	\$45.75
Active continuous queriers / MI300X	14.5
Bursty engineering team seats / MI300X	70-140
Owned MI300X capex	\$18,000 one-time

Lisa Su said: "AI is for everyone."

AMD CEO Lisa Su, CES 2026 keynote, Jan 5

We took it literally. **REPOMIND opens a \$5-10B market** that closed coding agents abandoned:

Hyperscalers · Banks · Defense · Pharma · Healthcare · Consumer Tech

Every team that can't or won't ship code to a SaaS coding agent.

AMD made the hardware. We made the open-source MIT unlock.